



AI & ML ENGINEERING TRACK

Curriculum & Project Roadmap

Recruitment-Mirrored Project-Based Learning

Four projects. One live defense. Every submission reviewed and approved by a working professional before it is credentialed.

Duration

7–8 weeks

Format

100% online, live

Cohort 1 begins

1 September 2026

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Why this track is built the way it is

Most technical courses assess completion. You watch the modules, submit the notebook, and a score is generated. The problem is structural rather than lazy: automated graders can only mark work that has one correct answer. So that is the work they set.

But almost nothing an engineer is actually paid to do has one correct answer. Choosing an evaluation metric, noticing that a dataset is leaking, deciding where a system should fail loudly instead of quietly — these are judgement calls. They cannot be auto-graded, so most curricula quietly leave them out, and candidates arrive at interviews having never made one.

This track is built the other way round. Every project sets a problem with more than one defensible answer, and a working professional reviews the reasoning, not just the output.

"You cannot certify yourself. A working professional mentor must start, review and approve your work."

Recruitment-Mirrored Project-Based Learning (RM-PBL)

Each of the four core projects is designed to mirror a specific stage of a real technical hiring funnel. The intent is that the assessment you sit here and the assessment you sit at a company are structurally the same exercise — so the track is preparation in the literal sense, not by analogy.

Project	Hiring stage it mirrors	Core judgement tested
Solo 1	Technical phone screen	Can you produce a result someone else can reproduce?
Solo 2	Applied domain round	Can you tell when a good number is a lie?
Pair	Live system design round	Can you agree an interface before writing code?
Capstone	Full funnel, end to end	Can you state what your own system gets wrong?

Who this track is for

- Engineering students in CSE, IT, AI & DS or adjacent branches, typically pre-final or final year
- Recent graduates targeting ML Engineer, AI Engineer, NLP Engineer or Data Scientist roles
- Working professionals moving into applied ML from software, data or analytics roles

What you need before you start. You should be comfortable reading and writing basic Python — functions, loops, lists and dictionaries. You do not need prior machine learning experience, statistics beyond school level, or a mathematics background. Library-specific work (pandas, scikit-learn, PyTorch) is taught inside the track.

What this track is not. It is not an introduction to programming, and it is not a survey course. The pace assumes you can already write code and want to learn to make engineering decisions with it.

The four-project spine

These four projects run the length of the track and carry the assessment. Each one is briefed like a work assignment rather than an exercise: you are given a situation and a constraint, not a set of steps.

01 Solo Project 1 — Baseline and reproducibility

Weeks 1–2

MIRRORS: TECHNICAL PHONE SCREEN

You are given a clean dataset and a specified target metric. Build and evaluate a baseline model. The deliverable is not the highest score — it is a result that runs again and produces the same numbers. Same seed, same split, same output.

What is being assessed

- Whether your pipeline is deterministic and your results are reproducible
- Whether your evaluation is set up correctly before any modelling begins
- Whether your code can be read and re-run by someone who did not write it

02 Solo Project 2 — Dirty data and hidden leakage

Weeks 3–4

MIRRORS: APPLIED DOMAIN ROUND

You are given a dirty dataset with target leakage deliberately planted in it, and no stated metric. You choose the metric and justify the choice. You find the leak. Finding the leak is most of the grade — a model reporting excellent performance is the symptom, not the success.

What is being assessed

- Whether you interrogate a suspiciously good result instead of accepting it
- Whether you can select and defend an evaluation metric for an unstated problem
- Whether you can trace a signal back through a pipeline to its source

03 Pair Project — Shared RAG service

Weeks 5–6

MIRRORS: LIVE ML SYSTEM DESIGN ROUND

Two students build one retrieval-augmented generation service. One owns retrieval; the other owns generation and evaluation. Neither may write code until both have agreed an interface and an evaluation set. The negotiation is part of the assessment, because in practice it always is.

What is being assessed

- Whether you can specify a contract between two components before building either
- Whether you can define what 'working' means before you claim it
- Whether you can integrate with someone else's design decisions without rewriting them

04 Capstone — Student-proposed applied ML system

Weeks 7–8

MIRRORS: FULL FUNNEL, END TO END

You propose and build an applied ML system of your own, with a real evaluation story. The central requirement is honesty about limits: you must be able to say what your system gets wrong, under what conditions, and with what measurement behind the claim.

What is being assessed

- Whether you can scope a problem to something achievable and worth doing
- Whether your evaluation reflects real failure modes rather than a convenient test set
- Whether you can present and defend the whole system to a practitioner who will probe it

Eight targeted mini-projects

Data structures and algorithms are taught through applied AI systems work rather than as abstract puzzles. Each mini-project implements a structure inside a component you would plausibly ship — so the same preparation serves both a DSA interview round and the engineering work itself.

Structure	Built as
Hashing	Deduplication and embedding-cache layer for repeated inference calls
Heaps	Top-k retrieval ranker for a RAG search feature
Linked list + hashmap	LRU cache for inference responses
Trees (BST / Trie)	Prefix and vocabulary search over an embedding index
Graphs (BFS / DFS)	Dependency resolution for a multi-step agent workflow
Graphs (weighted)	Cost-aware router across multiple model endpoints
Dynamic programming	Token-budget and context-window allocator
Arrays / sliding window	Streaming-token rate limiter for API calls

Technical stack

Language and data	Python, pandas, NumPy
Classical ML	scikit-learn, model selection and evaluation
Deep learning	PyTorch, TensorFlow / Keras
NLP	NLTK, spaCy, embeddings and vector stores
Serving and deployment	Flask, FastAPI, Streamlit
Environment and tooling	Jupyter, Git, Docker

Tools are taught in the context of the project that needs them. There is no separate tools module.

Week by week

Wk	Focus	Milestone
1	Python for ML, data handling, evaluation set up before modelling	Solo 1 begins
2	Baseline models, reproducibility, reading someone else's pipeline	Solo 1 due
3	Messy data, feature engineering, metric selection and justification	Solo 2 begins
4	Target leakage, validation strategy, interrogating a good result	Solo 2 due
5	Embeddings, retrieval, interface contracts and evaluation sets	Pair begins
6	Generation, RAG failure modes, evaluating a system you built	Pair due
7	Capstone scoping, deployment, measuring real failure conditions	Capstone begins
8	Capstone completion, PAT defense, portfolio publication	Capstone due · PAT

Live mentor-led sessions run through the week, with recorded material available for revision. Weekly milestones are checkpoints, not deadlines you meet alone — mentor review happens at each one.

Assessment and the PAT

The Practical Assessment Test (PAT) is the credential-bearing component of the track. It is not a quiz and it cannot be retaken by resubmission. It is a live oral defense of work you built, conducted by a practitioner who will question your decisions.

- Mentor verification. Every project is reviewed and approved by a working professional before it counts. Nothing is self-certified.
- Live defense. You present your capstone and answer questions on why you chose what you chose — including what you would do differently.
- Open scoring. You receive a scored breakdown, not a pass mark. The rubric is visible to you and to recruiters reviewing your profile.
- Recommendation letter. Issued by the reviewing mentor where the standard is met.

What you leave with

- Four completed, mentor-verified projects with working code and live deployments
- A verified portfolio profile structured for recruiter review — architecture, code quality and assessment scores, not a static resume
- A scored PAT record that a recruiter can check independently
- Eight applied DSA implementations relevant to both interviews and production work

What we do not promise

We do not guarantee placement, and we would be cautious of anyone who does. What we control is the quality of the work you produce and the credibility of the assessment behind it.

SkillCred is a new company and this is our first cohort. We think the approach is right, and the people teaching it have spent years doing this work — but we have no alumni outcomes to show you yet, and we would rather say so than imply otherwise. Judge the curriculum, sit in on a live session, and decide from there.

Try it before you decide

FREE LIVE SESSION

Build a RAG system in 90 minutes — and debug it like an engineer

Monday 17 August 2026 · 7:00 PM IST · 90 minutes · Online

We build a working retrieval-augmented generation system live, from an empty file. Then we break it four ways — the answer that spans two chunks, the question the corpus cannot answer, the retrieval that looks healthy while the output is wrong — and work out how you would have caught each one.

Every one of those failures passes an automated check. Catching them is what the job actually is. It is the same problem you would build with a partner in week five of the track.

Not a course preview. You will build a real thing and leave with the code.

Reserve a seat at skillcred.in/lp/ai-ml

Cohort 1 — AI & ML Engineering Track

Begins	1 September 2026
Duration	7–8 weeks, live and online
Includes	Live mentor sessions, four verified projects, PAT defense, verified portfolio

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